

Alexis Manche

Research Fellow · Computational Materials Chemistry

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PROFILE

Computational materials researcher working at the interface of solid-state chemistry, electronic-structure theory and energy materials. Research focuses on point defects, dopants, charge localisation, ion/electron transport and first-principles spectroscopy, with an emphasis on reproducible atomistic workflows and experimentally testable predictions.

RESEARCH EXPERIENCE

Research Fellow, School of Chemistry — University of Birmingham (2024–present)

Computational materials research in the Scanlon Materials Theory Group.

- Point-defect and dopant thermodynamics using VASP, doped and ShakeNBreak workflows.
- Ion migration, small-polaron localisation and coupled ionic–electronic transport calculations.
- Development of automated spectroscopy and scientific-software workflows for atomistic simulations.
- Research on sodium-ion cathodes, transition-metal oxides and mixed ionic–electronic conductors.

Visiting Researcher — University of California, Santa Barbara (2020)

Computational and spectroscopic research on battery materials in the Clément group.

EDUCATION

PhD, Chemistry / Energy Materials — University of St Andrews (2020–2024)

Thesis: *The Challenge of Polyanion Redox in Oxalates*. Combined synthesis, electrochemistry, spectroscopy and first-principles modelling of Li/Na-ion cathodes.

MSc, Physical Chemistry — University of Bordeaux (2018–2020)

BSc, Chemistry — University of Bordeaux (2017–2018)

CORE METHODS & SOFTWARE

Electronic structure	VASP; DFT+U; hybrid DFT; r2SCAN/meta-GGA; charge localisation; density of states
Defects	doped; ShakeNBreak; pymatgen; formation energies; charge states; chemical potentials
Kinetics & lattice dynamics	NEB; migration pathways; polaron hopping; phonons; Raman/IR workflows
Scientific computing	Python; Linux; HPC; Git/GitHub; workflow automation; reproducible analysis

SELECTED RESEARCH THEMES

- **Defect chemistry & doping:** thermodynamic site preference, charge states, compensation and defect populations.
- **Transport:** ion migration and localised electronic carriers in mixed ionic–electronic conductors.
- **Spectroscopy:** computational fingerprints of dopants/defects in Raman, IR and X-ray observables.
- **Battery materials:** sodium-ion and lithium-ion electrodes, layered oxides, polyanions and transition-metal oxides.

SELECTED PUBLICATIONS

2025 T. Song, A. Manche, et al.

Breaking the Durability–Power Trade-Off: Boron-Directed Faceted O₃ Cathodes for High-Rate Sodium-Ion Batteries. *Advanced Energy Materials*.

DOI: 10.1002/aenm.202505657

2024 G. Pramanik, A. Manche, et al.

NaLiFe(C₂O₄)₂: a polyanionic Li/Na-ion battery cathode exhibiting cationic and anionic redox. *Energy Storage Materials* 73, 103821.

DOI: 10.1016/j.ensm.2024.103821

2023 G. Pramanik, A. Manche, et al.

KFe(C₂O₄)F: a fluoro-oxalate cathode material for Li/Na-ion batteries. *ChemElectroChem* 10, e202300192.

DOI: 10.1002/celec.202300192

2023 E. E. Foley, et al., A. Manche, et al.

Polymorphism in Weberite Na₂Fe₂F₇ and its Effects on Electrochemical Properties as a Na-Ion Cathode. *Chemistry of Materials* 35, 3614–3627.

DOI: 10.1021/acs.chemmater.3c00233

2023 G. Pramanik, A. Manche, et al.

K₂Fe(C₂O₄)₂: an oxalate cathode for Li/Na ion batteries exhibiting a combination of multielectron cation and anion redox. *Chemistry of Materials*.

DOI: 10.1021/acs.chemmater.3c00063

2022 G. Pramanik, A. Manche, et al.

Exploiting anion and cation redox chemistry in lithium-rich perovskite oxalate: a novel next-generation Li/Na-ion battery electrode. *Dalton Transactions* 51, 12467–12475.

DOI: 10.1039/D2DT01447E

2021 E. E. Foley, et al., A. Manche, et al.

Probing reaction processes and reversibility in Earth-abundant Na₃FeF₆ for Na-ion batteries. *Physical Chemistry Chemical Physics* 23, 20052–20064.

DOI: 10.1039/D1CP02763H

SELECTED PROJECTS

Defect-to-spectroscopy workflows

Developing scientific software to connect dopant/point-defect models to Raman, IR and X-ray spectroscopic signatures.

Layered oxide cathodes for Na-ion batteries

Ab initio modelling of structural, electronic and transport descriptors; includes leadership of a Faraday Institution FUSE internship project.

Point-defect chemistry of battery materials

Systematic modelling of intrinsic defects and dopants to resolve site preference, charge compensation and degradation mechanisms.

Coupled ion–polaron transport

Atomistic treatment of charge localisation and migration in mixed ionic–electronic conductors.

PROFESSIONAL LINKS

GitHub: github.com/alex-manche

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University profile: University of Birmingham research portal

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